

1. Let R denote the cell radius. Find the distance between A and p_1 , and the distance between B and p_1 , shown in Figure 1.

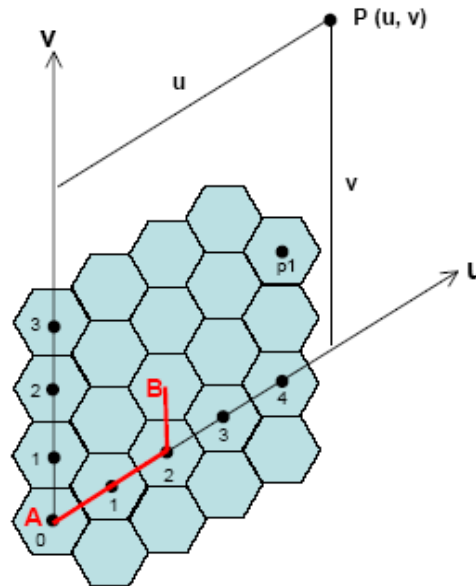


Figure 1 Hexagon geometry

2. In a cellular system, the received power at the mobile unit is -98 dBm. If the thermal noise power is -120 dBm and the co-channel interference (CCI) from each interfering station is -121 dBm, calculate the signal-to-CCI ratio and signal-to-noise ratio.
3. A service provider wants to deploy a cellular system with the required SIR of 24 dB.
 - (a) Determine the optimal frequency reuse factor N needed in this cellular system. Assume the path loss exponent $n=4$.
 - (b) If the service provider has a 25 MHz band and each user requires 30 KHz for voice communications. What would be the maximum number of simultaneous users (capacity) that the system could support in each cell?